

chapter 8

Reporting and Interpreting
Property, Plant, and Equipment;
Intangibles; and Natural
Resources

Financial Accounting

12e

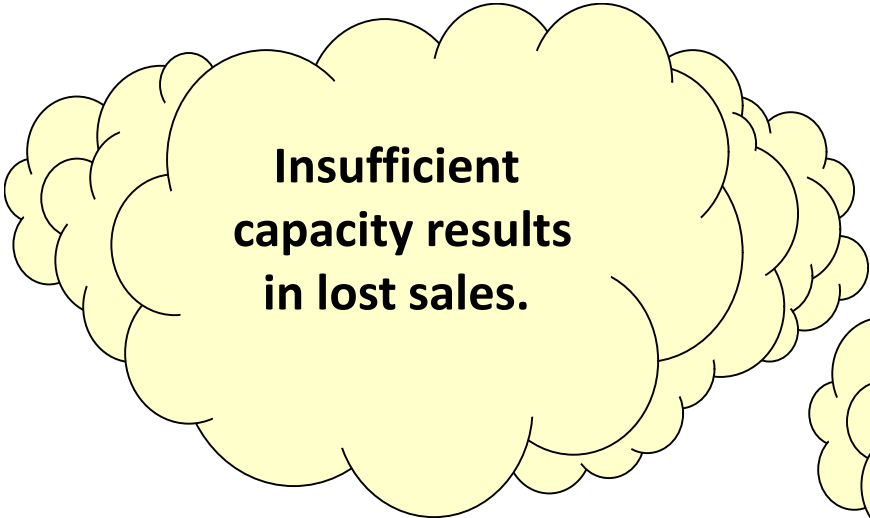
Libby • Libby • Hodge

Learning Objectives

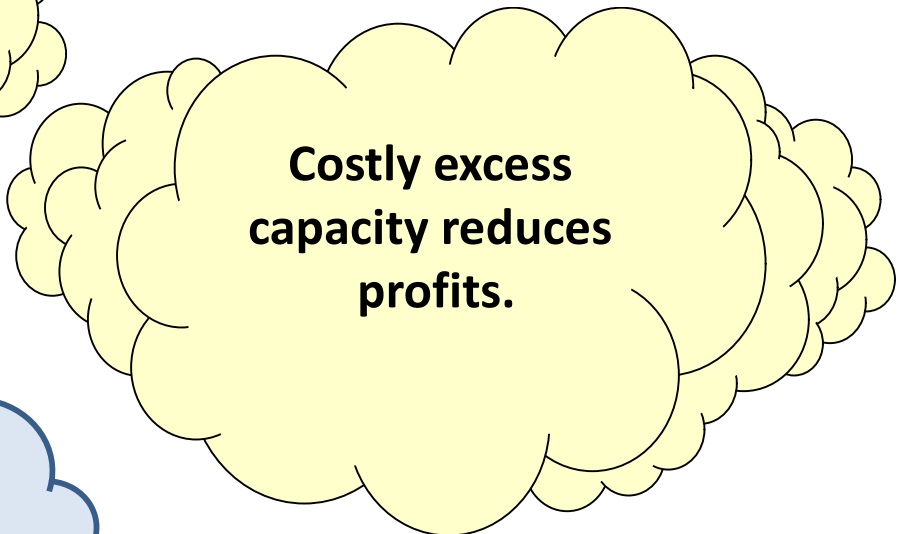
After studying this chapter, you should be able to:

- 8-1** Define, classify, and explain the nature of long-lived productive assets and interpret the fixed asset turnover ratio.
- 8-2** Apply the cost principle to measure the acquisition and maintenance of property, plant, and equipment.
- 8-3** Apply various cost allocation methods as assets are held and used over time.
- 8-4** Explain the effect of asset impairment on the financial statements.
- 8-5** Analyze the disposal of property, plant, and equipment.
- 8-6** Apply measurement and reporting concepts for intangible assets and natural resources.
- 8-7** Explain how the acquisition, use, and disposal of long-lived assets impact cash flows.

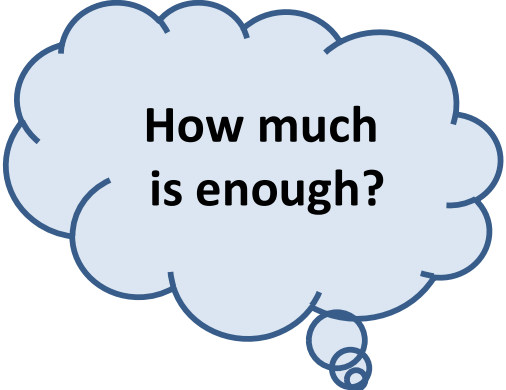
Understanding the Business



**Insufficient
capacity results
in lost sales.**



**Costly excess
capacity reduces
profits.**



**How much
is enough?**

Learning Objective 8-1

8-1 Define, classify, and explain the nature of long-lived productive assets and interpret the fixed asset turnover ratio.

Classifying Long-Lived Assets

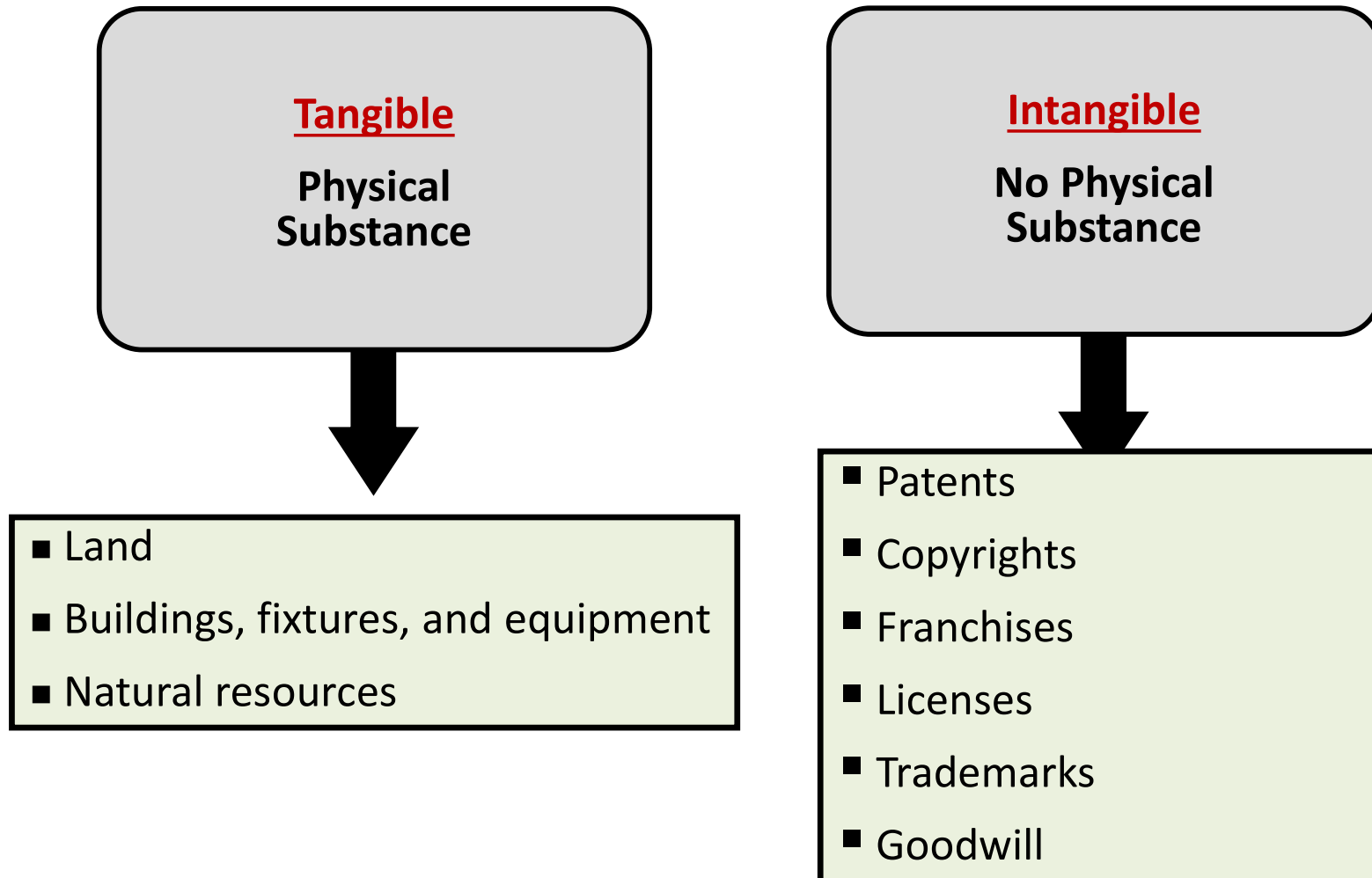


Exhibit 8.1 FedEx Corporation's Asset Section of the Balance Sheet

FEDEX CORPORATION Consolidated Balance Sheets (Partial) (in millions)		
	May 31	
	2023	2022
ASSETS		
CURRENT ASSETS <i>(summarized)</i>	\$ 18,610	\$20,365
PROPERTY AND EQUIPMENT, AT COST		
Aircraft and related equipment	29,108	27,874
Package handling and ground support equipment	16,839	14,930
Information technology	8,792	8,098
Vehicles and trailers	10,191	9,806
Facilities and other	15,694	14,567
Gross property and equipment	80,624	75,275
Less accumulated depreciation and amortization	39,926	37,184
Net property and equipment	40,698	38,091
OTHER LONG-TERM ASSETS		
Operating lease right-of-use assets, net	17,347	16,613
Goodwill	6,435	6,544
Other assets	4,053	4,381
Total other long-term assets	27,835	27,538
TOTAL ASSETS	\$ 87,143	\$85,994

Fixed Asset Turnover

KEY RATIO ANALYSIS



$$\text{Fixed Asset Turnover} = \frac{\text{Net Sales (or Operating Revenues)}}{\text{Average Net Fixed Assets}}$$

This ratio measures the sales dollars generated by each dollar of fixed assets used. A **high ratio** suggests **effective management**.

The FedEx ratio for fiscal year ended September 30, 2023 is (in millions):

$$\text{Operating Revenues } \$90,000 \div [(\$40,698 + \$40,698) \div 2] = 2.27 \text{ times}$$

COMPARISON OVER TIME		
FedEx Corporation		
	2022	2023
	2.42	2.27

COMPARISON WITH COMPETITOR	
United Parcel Service	
	2.95

Learning Objective 8-2

8-2 Apply the cost principle to measure the acquisition and maintenance of property, plant, and equipment.

Measuring and Recording Acquisition Cost (1 of 7)

Under the cost principle, all reasonable and necessary expenditures made in acquiring and preparing an asset for use (or sale, as in the case of inventory) should be recorded as the cost of the asset. *Interest charges associated with the purchase are recorded as expenses as incurred.*

Acquisition Costs - Buildings, Equipment, etc.

- Purchase price, sales taxes, legal fees, transportation costs, installation and preparation costs (including renovation and repairs)

Acquisition Costs - Land

- Purchase price, title fees and title insurance, sales commissions, legal fees, delinquent taxes, surveying fees.

Note - We say that the expenditures are **capitalized** when they are recorded as an asset, not as expenses in the current period.

Measuring and Recording Acquisition Cost (2 of 7)

The **Acquisition Cost** is:

- Net **cash amount paid** for the asset or
- The fair value of the asset given or asset received, whichever is more clearly determined (called the **cash equivalent price**), when noncash assets are used as payment.

For example, FedEx would calculate the acquisition cost of a new aircraft as follows:

Invoice price	\$26,800,000
Less: Discount from ATR	(1,800,000)
Net cash invoice price	<u>25,000,000</u>
Add: Transportation charges paid by FedEx	400,000
Preparation costs paid by FedEx	<u>600,000</u>
Cost of the aircraft (added to the asset account)	<u><u>\$26,000,000</u></u>

Measuring and Recording Acquisition Cost (3 of 7)

Basket Purchase: When land, buildings, and equipment are purchased as a group the total cost is allocated to each asset proportionately based on each asset's market value relative to the total market value of the assets as a whole

Building Cost = (Market value of building / Total market value of the basket purchase) × Total cost of basket purchase

Measuring and Recording Acquisition Cost (4 of 7)

Acquisition for Cash – Assume FedEx paid cash to purchase an aircraft for \$26 million, including related transportation and preparation costs.

	Debit	Credit
Aircraft (+A)	26	
Cash (-A)		26

Assets		=	Liabilities	+	Stockholders' Equity
Aircraft	+26				
Cash	-26				

Measuring and Recording Acquisition Cost (5 of 7)

Acquisition for Debt – Assume FedEx signed a note payable for the new aircraft and paid cash for the transportation and preparation costs.

	Debit	Credit
Aircraft (+A)	26	
Cash (-A)		1
Note Payable (+L)		25

Assets		=	Liabilities		+	Stockholders' Equity
Aircraft	+26		Note payable	+25		
Cash	-1					

Measuring and Recording Acquisition Cost (6 of 7)

Leasing Long-Term Assets:

- Instead of a note payable, companies may choose to **lease** long-term assets.
- Companies are required to report most types of leases on the balance sheet as **Operating Lease Assets**, with the related liabilities reported on the balance sheet as **Operating Lease Liabilities**.
- Because leases confer right-of-use to the company using the asset, these are considered **intangible assets** and are not included in property, plant, and equipment (fixed assets).
- The lessee (the one using the asset) will recognize:
 - depreciation on right-of-use assets and
 - interest on lease liability.



Companies widely use leases for office equipment and shopping carts, vehicles, farm equipment, and airplanes. Beginning in 2019 as required by new accounting standards, many companies, especially airlines and retailers, reported significant increases in assets and debt due to leases. **Remember:** Don't include the right-of-use operating lease assets in total fixed assets; they are intangible assets.

Measuring and Recording Acquisition Cost (7 of 7)

Acquisition for Equity - FedEx gave ATR 1,000,000 shares of new \$1.00 par value common stock with a market value of \$10 per share and paid the balance of the plane plus transportation and preparation costs in cash.

	Debit	Credit
Aircraft (+A)	26	
Common Stock (+SE) ¹		1
Additional paid-in capital (+SE) ²		9
Cash (–A)		16

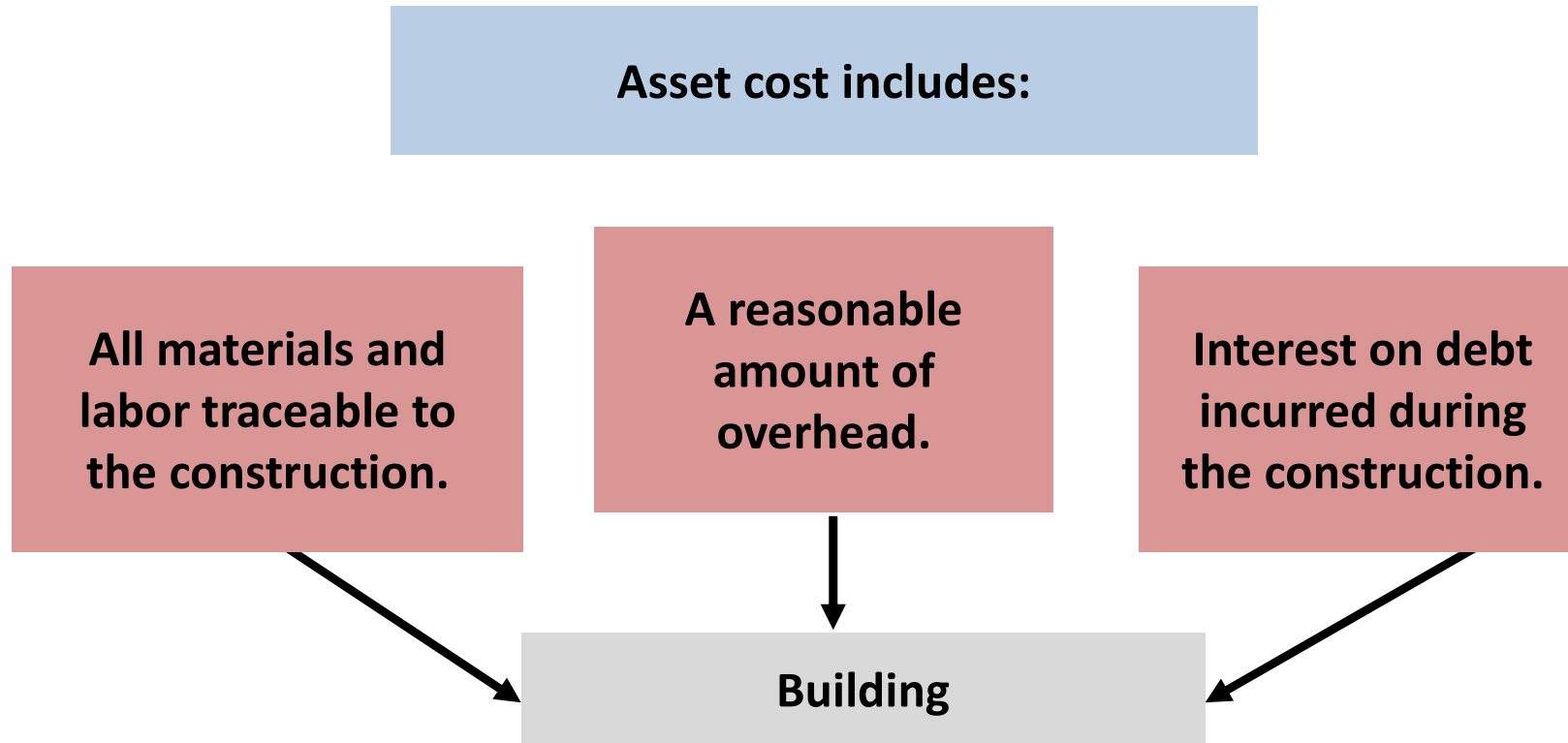
Assets		=	Liabilities	+	Stockholders' Equity	
Aircraft	+26				Common stock	+1
Cash	–16				Additional paid-in capital	+9

¹ 1,000,000 shares × \$1 par

² 1,000,000 shares × \$9 excess (\$10 market value – \$1 par)

Measuring and Recording Acquisition Cost

Acquisition by Construction (1 of 2)



Measuring and Recording Acquisition Cost Acquisition by Construction (2 of 2)

Assume FedEx constructed a new hangar, paying \$8 million in labor costs and \$11 million in supplies and materials. FedEx also paid \$1 million in interest expense during the year related to the construction project.

Capitalized Expenditures:

<i>Wages paid</i>	<i>\$ 8</i>
<i>Materials and supplies paid</i>	<i>11</i>
<i>Interest paid</i>	<i>1</i>



Remember: All reasonable and necessary expenditures made in acquiring and preparing an asset for its intended use should be recorded as part of the cost of the asset—they should be **capitalized**! But, after the asset is ready to use, no other costs, such as insurance on the asset while in use, are capitalized.

The following journal entry is made to record the asset:

	Debit	Credit
Facilities (+A)	20	
Cash (−A)		20

Assets		=	Liabilities	+	Stockholders' Equity
Facilities	+20				
Cash	−20				

Repairs, Maintenance, and Improvements (1 of 3)

Type of Expenditure	Identifying Characteristics	Accounting Treatment
Ordinary repairs and maintenance	1. Maintains the productive capacity of the asset during the current accounting period only 2. Recurring in nature 3. Involve relatively small amounts 4. Do not increase the productive life, operating efficiency, or capacity of the asset	Expense in the period incurred
Improvements	1. Increase the productive life, operating efficiency, or capacity of the asset 2. Occur infrequently 3. Involve large amounts of money	Add to asset account (Capitalize)

Repairs, Maintenance, and Improvements (2 of 3)

Treatment	Financial Statement Effect			
	Statement	Expense	Current Income	Current Taxes
Capitalize	Balance sheet account debited	Deferred	Higher	Higher
Expense	Income statement account debited	Currently recognized	Lower	Lower

To save time, some companies record all expenditures below a certain dollar amount as expenses. Such policies are acceptable because immaterial amounts will not affect users' decisions when analyzing financial statements.

Capitalizing expenses will increase assets and net income in the current year, lowering future years' income by the amount of the annual depreciation.

Repairs, Maintenance, and Improvements (3 of 3)

Repairs and maintenance: FedEx reported \$3,357 million in Maintenance and Repairs on its 2023 income statement.

	Debit	Credit
Maintenance and repairs expense (+E, -SE)	3,357	
Cash (-A)		3,357

Assets		=	Liabilities	+	Stockholders' Equity	
Cash	-3,357				Maintenance and repairs expense (+E)	-3,357

Improvements: Assume FedEx spent \$300 million to modify the cargo area of its oldest aircraft to reduce weight, resulting in 9 percent greater fuel efficiency and lower operating costs. These expenditures would have been recorded as capital expenditures:

	Debit	Credit
Aircraft (+A)	300	
Cash (-A)		300

Assets		=	Liabilities	+	Stockholders' Equity	
Aircraft	+300					
Cash	-300					

WorldCom: Hiding Billions in Expenses through Capitalization

A QUESTION OF ETHICS



- When expenditures that should be recorded as current period expenses are improperly capitalized as part of the cost of the asset, the effects on the financial statements can be enormous.
- In one of the largest accounting frauds in history, **WorldCom** (now part of Verizon) **inflated its income and cash flows from operations by billions of dollars**. This fraud turned WorldCom's actual losses into large profits.
- Over five quarters in 2001 and 2002, the company initially announced that it had capitalized \$3.8 billion that should have been recorded as operating expenses. By early 2004, auditors had discovered **\$11 billion in necessary restatements** (reductions to previously reported pretax income) for 2000 and 2001.
- Accounting for expenses as capital expenditures increases current income because it spreads a single period's operating expenses over many future periods as depreciation expense. It increases cash flows from operations by moving cash outflows from the operating section to the investing section of the cash flow statement.

Being a Global Citizen

ENVIRONMENTAL, SOCIAL, & GOVERNANCE (ESG) REPORTING

FedEx published its ESG report at [fedex.com/en-us/sustainability/reports.html](https://www.fedex.com/en-us/sustainability/reports.html) stating that “ESG is integrated into our culture, and we align our sustainability approach with our core values, and long-term strategy. Our ESG Report details the company’s progress toward meeting its environmental, social, and governance strategies, goals, and initiatives. Here is a snippet from the ESG Report:

Fuel- and energy-saving initiatives		FY21	FY22
Energy saved (terajoules)			
FedEx Express aircraft fleet modernization		20,783	21,458
FedEx Fuel Sense (jet fuel) ^[1]	10,100	9,273	1,844
FedEx Express Vehicle fuel efficiency ^[3]	4,476	2,005	5,040
Intermodal rail usage ^[2]	3,324		9,049
Facility energy efficiency ^[4]	985		813

¹ FY21 and FY22 Fuel Sense savings were impacted by the significant increase in shipping volume and activity due to the COVID-19 pandemic.

² FY21 and FY22 include fuel and emission savings from intermodal rail usage at FedEx Freight Ground. FY20 does not include intermodal rail usage for FedEx Freight as data was not available for FedEx Ground prior to FY20.

³ FedEx Express vehicle efficiency initiatives include building lighting retrofits and energy management systems.

⁴ Facility energy efficiency savings were calculated using the EPA GHG Calculator.

Securing Fixed Assets

It is **management's responsibility** to establish internal controls to **secure long-lived assets**. Examples of these controls include:

- Creating a register (or ledger) of each fixed asset with identifying information, such as the purchase date, serial number, acquisition date, and cost.
- Physically "tag" each asset with an identifying technology, such as QR codes, simply this task.
- Compare the registry of assets to physical locations on an annual basis.
- Establishing and following policies for acquisition, depreciation, and approval procedures for acquisition and disposal of assets.

The controls surrounding the disposal of assets are often lacking. It is important that management establish internal control procedures to **secure cash** (that is a different period) at each point of sale, such as: approving the sale, recording the sale in the accounting system, reconciling the accounting records, and physically possessing the cash from the sale. Otherwise, there is a risk of loss to the company.



Tip

Reminder: Establishing internal controls is the responsibility of management. Auditors have the responsibility to test the internal controls for weaknesses that could affect financial information.

Learning Objective 8-3

8-3 Apply various cost allocation methods as assets are held and used over time.

Depreciation Concepts (1 of 2)

Long-lived assets, *except for land*, represents the prepaid cost of future services or benefits.

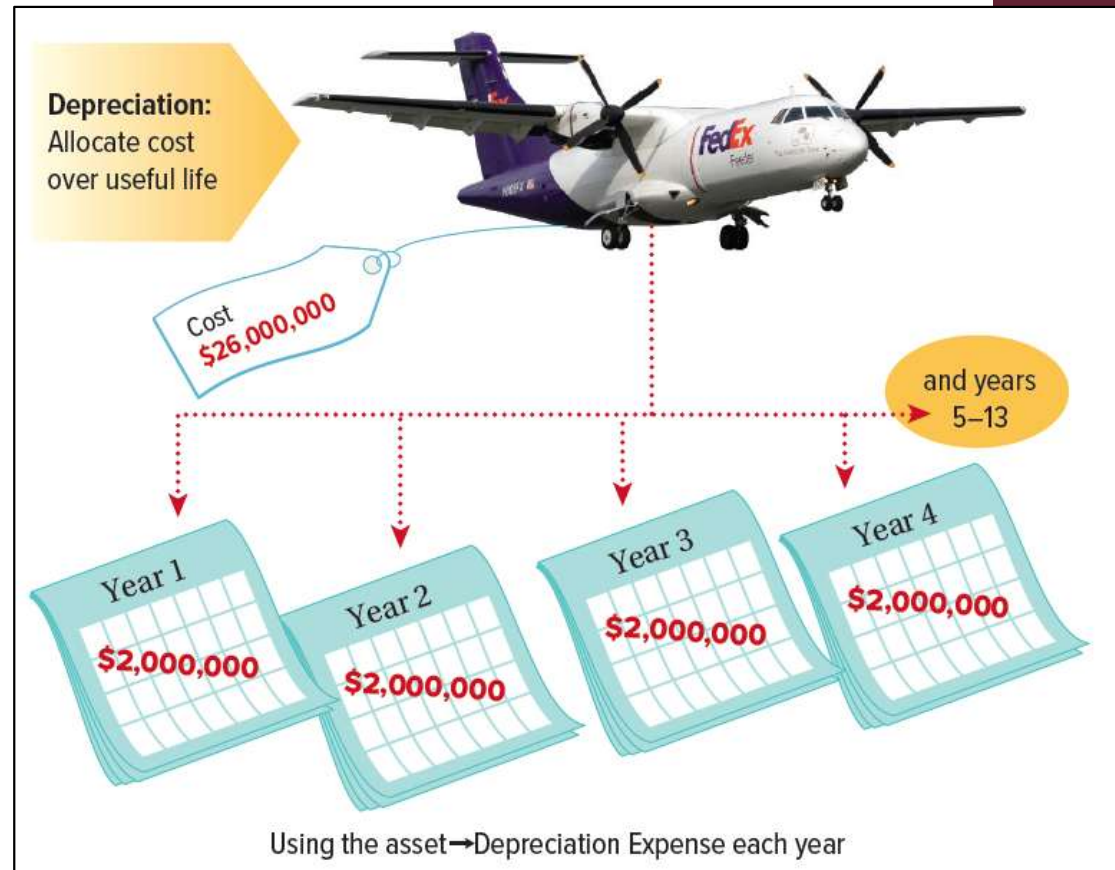
The **expense recognition (matching) principle** requires that a portion of an asset's cost be allocated as an expense in the same period that revenues are generated by its use.

The term used to identify the matching of the costs of using buildings and equipment with the revenues they generate is called **depreciation**.



Depreciation Concepts (2 of 2)

- ❖ Depreciation is the process of **allocating the cost** of buildings and equipment over their productive lives using a **systematic and rational method**.
- ❖ Depreciation is **not** a process of **determining** an asset's current **market value** or worth. The remaining balance sheet amount probably does not represent the asset's current market value.



Depreciation has nothing to do with an asset's market value (as if you were going to sell it) because **your intent is to use it**.

Adjusting for Depreciation (1 of 4)

An adjusting journal entry is needed at the end of each period to reflect the use of buildings and equipment for the period.

	Debit	Credit
Depreciation expense (+E, -SE)	x,xxx	
Accumulated depreciation (+XA, -A)		x,xxx

Assets	=	Liabilities	+	Stockholders' Equity
Accumulated depreciation (+XA) -x,xxx				Depreciation expense (+E) -x,xxx

Cost	XX
<u>- Accumulated Depreciation</u>	<u>- XX</u>
Net Book Value	XX



Tip Again → In accounting, depreciation is simply a **cost allocation** concept—net book value is reduced as the asset is used to generate revenue. It has nothing to do with an asset's market value (as if you were going to sell it) because your intent is to use it.

Adjusting for Depreciation (2 of 4)

- The amount of depreciation recorded during each period is reported on the income statement as **Depreciation Expense**.
- The amount of depreciation expense accumulated since the acquisition date is reported on the balance sheet as a contra-account, **Accumulated Depreciation**, and deducted from the related asset's cost.
- The net amount on the balance sheet is called **net book value** or **carrying value**.



On financial statements:

Income Statement

Depreciation expense: **Estimated cost used
in current period**

Balance Sheet

Property, Plant, and Equipment (Cost)	– Accumulated Depreciation (Used cost)	= Net Book Value (Unused cost)
--	---	---

Adjusting for Depreciation (3 of 4)

Book Value:
Cost less
accumulated
depreciation

Cost
\$26,000,000



	Year 1	Year 2	Year 3	Year 13
Cost	\$26,000,000	\$26,000,000	\$26,000,000	\$26,000,000
– Accumulated depreciation	2,000,000	4,000,000	6,000,000	26,000,000
Net book value	<u>\$24,000,000</u>	<u>\$22,000,000</u>	<u>\$20,000,000</u>	<u>\$ 0</u>

Reported on the
Balance Sheet

Adjusting for Depreciation (4 of 4)

Once an asset's cost is determined, to calculate depreciation expense, three pieces of information are required for each asset:

1. **Estimated useful life** to the company
2. **Estimated residual** (or salvage) value at the end of the assets' useful life to the company
3. **Depreciation method**

***Note:** Estimated useful life represents management's estimate of the asset's **useful economic life to the company** rather than its total economic life to all potential users.*



Notice that, since the asset's useful life and residual values are estimates, **depreciation expense is an estimate**. Companies ask equipment suppliers what the entire expected life is (say 20 years). Then, management decides, based on the company's own experience with similar assets and policies on equipment renewal, as to the estimated life and residual value to assign (say, 15 years with a 10 percent residual value at the end).

Managers may choose from these depreciation methods:

- ① Straight-line
- ② Units-of-production
- ③ Declining-balance

Book Value as an Approximation of Remaining Life

FINANCIAL ANALYSIS



Some analysts compare the book value of assets to their original cost as an approximation of their remaining life.

Example:

- If book value of an asset is 100 percent of its cost, it is a new asset.
- If book value of an asset is 25 percent of its cost, it only has around 25 percent of its life remaining.

***FedEx's** book value of its property and equipment (\$40,698 million) was nearly 51 percent of its original cost (\$80,624 million) on May 31, 2023, compared to nearly 52 percent for **United Parcel Service** and 57 percent for German company **Deutsche Post DHL Group**.*

- This comparison suggests that the flight equipment of FedEx and UPS may be **slightly older** than the equipment at DHL.
- This comparison is only a rough approximation and is **influenced by accounting issues such as how fast a company depreciates its assets.**

Differences in Estimated Lives within a Single Industry

FINANCIAL ANALYSIS

	Estimated Useful Life
FedEx: Wide-body	15 to 20
Narrow-body	10 to 15
United Parcel Service	10 to 15
Deutsche Post DHL Group	10 to 25

- Differences in estimated lives may be attributable to a number of factors such as the type of equipment used by each company, equipment replacement and operation practices, and the degree of management's conservatism.
- Also, companies that plan to use the equipment for many years may estimate higher residual values.
- Differences in estimated lives and residual values of assets may have a significant impact on a comparison of the profitability of the companies.

Exhibit 8.2 Data for Illustrating the Computation of Depreciation under Alternative Methods

Assume that **ABC Company** acquired a new delivery van on June 1, 2026, the start of the fiscal year.

Acquisition of Delivery Van #16335PA

Cost, purchased on June 1, 2026	\$62,500
Estimated residual value	\$ 2,500
Estimated useful life	3 years OR 100,000 miles
Actual miles driven in:	
Year ended May 31, 2027	30,000 miles
Year ended May 31, 2028	50,000 miles
Year ended May 31, 2029	20,000 miles

Straight-Line Method (1 of 2)

Straight-Line Formula:

$$(\text{Cost} - \text{Residual Value}) \times 1/\text{Useful Life} = \text{Depreciation Expense}$$

Depreciable Cost

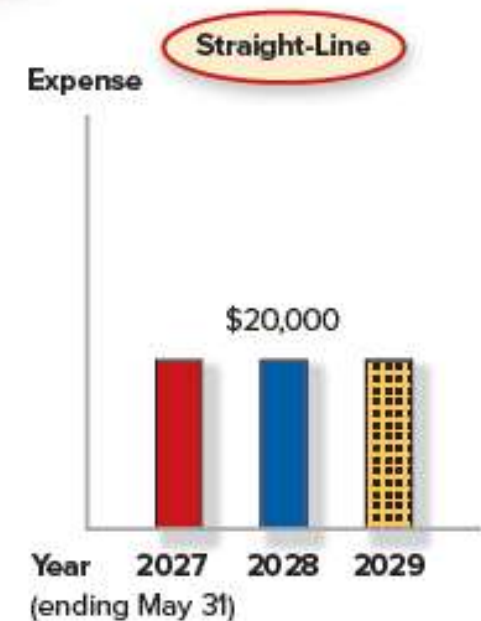
Straight-Line Rate

Notice that:

- ❖ Most companies select the straight-line depreciation method because it is the simplest method.
- ❖ The straight-line method is used when no other cost allocation method provides a better matching of revenues and expenses.
- ❖ Depreciation expense is a constant amount each year.
- ❖ Accumulated depreciation increases by an equal amount each year.
- ❖ Net book value decreases by the same amount each year until it equals the estimated residual value.

Straight-Line Method (2 of 2)

File Home Insert Page Layout Formulas Data Review View Help Power Pivot					
E25					
	A	B	Reported on the Income Statement (closed at year-end)		E
1					
2		Straight-Line Method:			Delivery Van #91-DHH16335PA
3			Depreciation	Accumulated	Net
4	Year	Computation	Expense	Depreciation	Book Value
5	Ending	(C-RV) x 1/UL	(for A/E)	(Balance after A/E)	(C-AD)
6	6/1/2026	Cost at acquisition		\$0	\$62,500
7	5/31/2027	$(\$62,500 - \$2,500) \times 1/3$	\$20,000	\$20,000	\$42,500
8	5/31/2028	$(\$62,500 - \$2,500) \times 1/3$	\$20,000	\$40,000	\$22,500
9	5/31/2029	$(\$62,500 - \$2,500) \times 1/3$	\$20,000	\$60,000	\$2,500
10					
11					
12					
13					



Units-of-Production Method (1 of 3)

- The **units-of-production depreciation** method relates depreciable cost to total estimated productive output or activity level.
- It can be used for depreciating equipment when production or activity levels vary over time, generating varying amounts of revenue—a good matching of expenses with revenues.
- The units-of-production method is used when activity or production level is the best determinant of life (e.g. commonly used for depreciating and depleting nature resources and related property).

Units-of-Production Method (2 of 3)

Acquisition of New Delivery Van #91-DHH16335PA

Cost, purchased on June 1, 2023	\$62,500
Estimated residual value	\$ 2,500
Estimated useful life	3 years OR 100,000 miles
Actual miles driven in:	
Year ended May 31, 2024	30,000 miles
Year ended May 31, 2025	50,000 miles
Year ended May 31, 2026	20,000 miles

Units-of-Production Formula:

$$\frac{(\text{Cost} - \text{Residual Value})}{\text{Estimated total production (or activity level) units}}$$

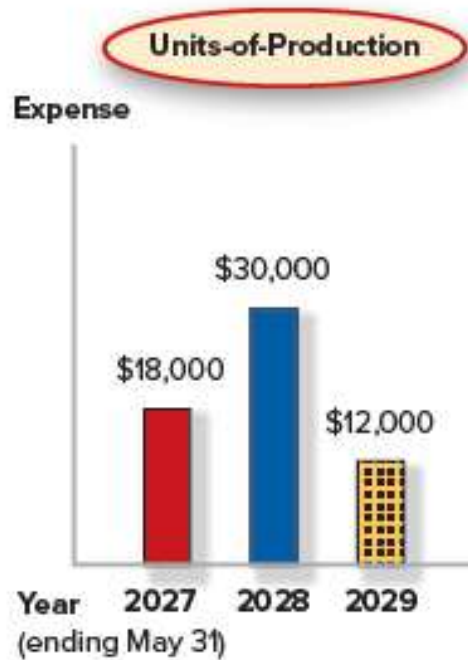
$$\text{Unit Rate} \times \text{Actual production or activity level for the period} = \text{Depreciation Expense}$$

First, divide the depreciable cost (cost minus residual value) by the estimated total production or activity level to determine the depreciation **unit rate**.

$(C - RV) / \text{Est total prod} = \text{Unit Rate}$
$(\$62,500 - \$2,500) / 100,000 \text{ miles} = \$0.60 \text{ per mile Unit Rate}$

Then multiply the unit rate by the actual production or activity level for the period to determine the period's depreciation expense.

Units-of-Production Method (3 of 3)



File	Home	Insert	Page Layout	Formulas	Data	Review	View	Help	Power Pivot
B10									
	A	B	C	D	E	F	G	H	
1									
2		Units-of-Production Method:				Delivery Van #91-DHH16335PA			
3		(C-RV)/Est. total activity =			Unit Rate				
4		(\$62,500 - \$2,500)/100,000 miles =			\$0.60	Depreciation	Accumulated	Net	
5			Computation			Expense	Depreciation	Book Value	
6		Year ending	Unit Rate x Actual Activity			(for A/E)	(Balance after A/E)	(C-AD)	
7		6/1/2026	Cost at acquisition				\$0	\$62,500	
8		5/31/2027	\$0.60	x 30,000 miles		\$18,000	\$18,000	\$44,500	
9		5/31/2028	\$0.60	x 50,000 miles		\$30,000	\$48,000	\$14,500	
10		5/31/2029	\$0.60	x 20,000 miles		\$12,000	\$60,000	\$2,500	
11									
12									
13									

Equal to estimated Residual Value at end of useful life

Declining-Balance Method

Accelerated Depreciation Method

If an asset is more efficient or productive when it is new, managers may choose the **declining-balance depreciation method** to recognize higher depreciation expense in the earlier years of an asset's life and lower depreciation expense in the later years.

This is an **accelerated** depreciation method.
Accelerated methods are selected for financial reporting

	<u>Depreciation Expense</u>	<u>Revenues</u>
Early Years	High	High
Late Years	Low	Low

Double Declining-Balance Method (1 of 3)

Double Declining-Balance Formula:

$$(\text{Cost} - \text{Accumulated Depreciation}) \times 2/\text{Useful Life} = \text{Depreciation Expense}$$

Net Book Value
(decreases over time)

Double the
straight-line rate

Annual computation of depreciation expense and annual value.

Tip

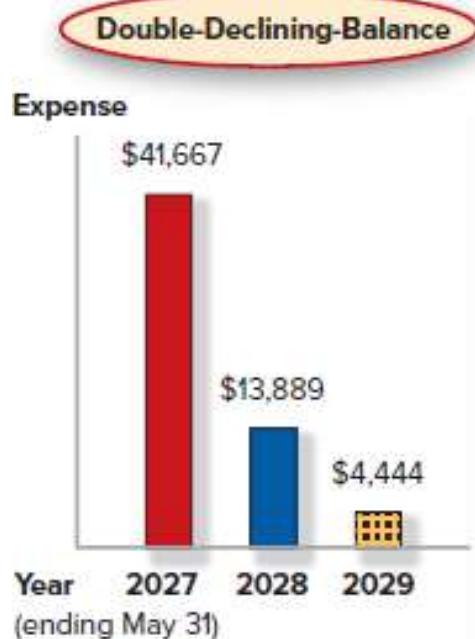
Unlike other methods, the double declining-balance formula calculates depreciation expense into perpetuity—net book value decreases each period, but never reaches \$0. Check each period that the net book value does not drop below residual value!

Double-Declining-Balance Method (2 of 3)

Two important differences between this method and the straight-line method:

1. Accumulated depreciation, not residual value, is included in the formula.
 - Because accumulated depreciation increases each year, net book value (Cost minus Accumulated depreciation) decreases.
 - **The double-declining-balance method applies a lower net book value each year, resulting in a higher depreciation expense over time.**
2. As with the other methods, **net book value should not be depreciated below the residual value.**
 - If the computation results in net book value below residual value, only the amount of depreciation expense needed to make net book value equal to residual value is recorded. Any additional depreciation expense is computed in subsequent periods.
 - In the final year of the asset's estimated useful life, an adjustment is needed to bring net book value to residual value. This adjustment is recorded, regardless of the amount of the depreciation computation.

Double Declining-Balance Method (3 of 3)



File Home Insert Page Layout Formulas Data Review View Help					
	B	C	D	E	F
2		Double Declining-Balance Method			Asset #91-DHH16335PA
3				Accumulated	Net
4		Computation	Expense	Depreciation	Book Value
5	Ending	(AD) x	(for AJE)	(Balance after AJE)	(C-AD)
6	6/1/2026			\$0	\$62,500
7	5/31/2027	1/3	\$41,667	\$41,667	\$20,833
8	5/31/2028	2/3	\$13,889	\$55,556	\$6,944
9	5/31/2029	3/3	\$4,444	\$60,000	\$2,500
10					
11					
12					

1. In the last year of asset's useful life, set Net Book Value equal to Residual Value. Then work backward.

2. Set Accumulated Depreciation equal to depreciable cost minus prior Accumulated Depreciation balance.

Depreciation expense is limited to the amount that reduces book value to the estimated residual value.

Summary of the Three Depreciation Methods

Method	Formula
Straight-line	$(\text{Cost} - \text{Residual Value}) \times 1/\text{Useful Life}$
Units-of-production	$(\text{Cost} - \text{Residual Value})/\text{Estimated Total Production} = \text{Unit Rate}$ $\times \text{Annual Production}$

Method	Pattern of Depreciation Expense
Straight-line	Equal amounts each year
Units-of-production	Varying amounts based on production level

Impact of Alternative Depreciation Methods

FINANCIAL ANALYSIS

Accelerated depreciation methods report higher depreciation expense during the early years of an asset's life. As the age of the asset increases, this effect reverses.

Summary Depreciation Expense



The graph illustrates the difference between the double-declining-balance method and the straight-line method of depreciation.

Differences in depreciation methods rather than real economic differences can cause significant variation in reported net income.

Increased Profitability Due to an Accounting Adjustment?

Reading the Notes

FINANCIAL ANALYSIS



- Financial analysts are interested in changes in accounting estimates because they impact a company's before-tax operating income.
- In 2023, **Alphabet** (the parent company of **Google**) and **IBM** disclosed their move to lengthen the depreciable lives of their server and network equipment, a move that boosted their income.
 - Alphabet's change reduced the company's depreciation expense by \$3.9 billion, resulting in a \$3.0 billion increase in net income.
 - IBM cited advances in technology for its change in useful life estimates, causing an expected pretax profit of over \$200 million.
- Analysts pay close attention to a change in accounting estimates because it represents increased profitability due merely to an accounting adjustment.

Objectives of GAAP versus Tax Reporting

Financial Reporting (GAAP)	Tax Reporting (IRC)
The objective of financial reporting is to provide economic information about the business that is useful in projecting future cash flows of the business. Financial reporting rules follow generally accepted accounting principles.	The objective of the Internal Revenue Code is to raise sufficient revenues for the expenses of the federal government. Many of the Code's provisions are designed to encourage certain behaviors that are thought to benefit society.
Most companies maintain two sets of accounting records. One set is for the stockholders and one for tax reporting.	

How Managers Choose

Financial Reporting (GAAP)	Tax Reporting
Managers determine which depreciation method provides the best matching of revenues and expenses. - Choose the straight-line method (the most common and easy to use) if the asset provides benefits evenly over time. Managers find this method the easiest to use and to explain. - Choose an accelerated method if assets produce more revenue in their early lives.	When a choice exists, managers will apply the latest and latest rule. All managers want to pay the lowest amount of tax that is legally permitted and at the latest possible date.

Two Sets of Books?

A QUESTION OF ETHICS



- It is both legal and ethical to maintain separate records for tax and financial reporting purposes. However, these records must reflect the same transaction.
- Understating revenues or overstating expenses on a tax return can result in financial penalties and imprisonment.
- Accountants who aid taxpayers can be fined or imprisoned and lose their professional licenses.

Deferred Tax Liabilities

By maintaining thousands of books, corporations can defer (delay) paying millions and some billions of dollars in taxes.

The following companies reported significant gross deferred tax obligations (liabilities) in their most recent annual reports. Much of these deferrals were due to differences in their choice of asset cost allocation (depreciation) methods.

Company	Deferred Tax Liability	Percentage Due to Applying Different Depreciation Methods
FedEx Corporation	\$ 1,027 million	99%
General Motors Corporation	\$ 4,233 million	86
Amazon.com	\$ 9,039 million	33
Costco Wholesale Corporation	\$ 1,989 million	44

Learning Objective 8-4

8-4 Explain the effect of asset impairment on the financial statements.

Measuring Asset Impairment

Corporations must review long-lived tangible and most intangible assets for possible **impairment** when events or changed circumstances cause the estimated future cash flows of these assets to fall below their book value

There are **three steps** to test for impairment:

Step 1: Consider whether **indicators** of impairment are present.

Step 2: **Test the recoverability** of the long-lived assets.

Step 3: **Measure the impairment** of the long-lived assets

Measuring Asset Impairment

There are **three steps** to test for impairment of long-lived tangible and most intangible assets. This is step 1:

Step 1: Consider whether indicators of impairment are present.

Companies must look for indicators or triggers to decide when to impair assets. The following factors may indicate impairment of assets:

- Significant pattern of decline of market price of the assets
- Lower demand due to global economic conditions
- Change in economic or legal factors (e.g., regulatory or technology changes)
- Obsolescence or physical damage to the asset
- Asset is held for disposal or is part of a restructuring

Measuring Asset Impairment

There are **three steps** to test for impairment. Here are steps 2 & 3:

Step 2: Test the recoverability of the long-lived assets.

When indicators of impairment exist, management assesses if the asset's net book value is recoverable through estimated future cash flows. The asset is **impaired if not recoverable** when:

Net book value (NBV) > Estimated future cash flows (FCF)

Step 3: Measure the impairment of the long-lived assets.

For any asset considered to be impaired, companies recognize a loss for the difference between the asset's net book value and its **fair value** (a market concept). The impaired asset is **written down** to fair value.

Net book value (NBV) – Fair Value (FV) = Impairment loss

A journal entry is made to record the loss due to impairment and reduce the asset. The fair value becomes the new book value. *The asset is not written back up if conditions change in the future.*

Tip

Estimating future cash flows is subjective and requires management to use significant judgment, basing future cash flow projections on reasonable and supportable assumptions for conditions that will exist over the remaining useful life of the asset. External auditors will review the assumptions for reasonableness.

Tip

After determining that the indicators suggest asset impairment

Step 1, the three numbers needed are:



Asset Impairment Illustration

Delta Air Lines reported the following in its 2020 10-K report (Note 2).
Dollars are in millions:

Step 1 → Indicators: “The unprecedented, widespread and persistent impact of COVID-19 and the related travel restrictions and social distancing measures implemented throughout the world have significantly reduced demand for air travel. . . (T)he spread of the virus and the resulting global pandemic have significantly affected our entire network.”

Step 2 → Recoverability: 383 aircraft were affected.

Net book value \$4,909 > Estimated future cash flows \$900 (assumed)

The assets are **impaired**.

Step 3 → Impairment measurement: The aircraft were retired from service.

Net book value \$4,909 – Fair value \$500 = **\$4,409 impairment loss**

	Debit	Credit
Loss due to impairment (+E, –SE)	4,409	
Aircraft (–A)		4,409

Assets		=	Liabilities	+	Stockholders' Equity	
Aircraft	–4,409				Loss due to impairment (+E)	–4,409

Learning Objective 8-5

8-5 Analyze the disposal of property, plant, and equipment.

Disposal of Property, Plant, and Equipment (1 of 5)

Businesses dispose of assets:

- **voluntarily** (sales, trade-ins, and retirements) or
- **involuntarily** as the result of a casualty such as a storm, fire, theft, or accident)

The asset disposal rarely happens at the end of the year therefore two journal entries are required as follows:

❶ Update depreciation expense through the date of disposal



❷ Record disposal by:

- Recording **cash** received (debit) or paid (credit)
- Writing off the **asset cost** (credit) and any **accumulated depreciation** (debit).
- Recording a **gain** (credit) or **loss** (debit).

Disposal of Property, Plant, and Equipment (2 of 5)

FedEx sold an aircraft for \$11 million cash at the end of its 17th year of use. The aircraft originally cost \$30 million and was depreciated using the straight-line method with zero residual value and a useful life of 25 years.

Calculate the annual depreciation expense:

- a. \$0
- ☒ b. \$1.2 million
- c. \$1.5 million
- d. \$2 million

Annual Depreciation:
 $(\$30,000,000 - \$0) \div 25 \text{ years}$
 $= \$1,200,000$

Disposal of Property, Plant, and Equipment (3 of 5)

FedEx sold an aircraft for \$11 million cash at the end of its 17th year of use. The aircraft originally cost \$30 million and was depreciated using the straight-line method with zero residual value and a useful life of 25 years.

After updating the depreciation, the equipment's book value at the end of the 17th year is:

- a.** \$9.6 million
- b. \$20.4 million
- c. \$12.8 million
- d. \$6.6 million

Accumulated Depreciation =

$$(17\text{yrs.} \times \$1,200,000) = \$20,400,000$$

BV = Cost – Accumulated Depreciation

$$\begin{aligned}\text{BV} &= \$30,000,000 - \$20,400,000 \\ &= \$9,600,000\end{aligned}$$

Disposal of Property, Plant, and Equipment (4 of 5)

FedEx sold an aircraft for \$11 million cash at the end of its 17th year of use. The aircraft originally cost \$30 million and was depreciated using the straight-line method with zero residual value and a useful life of 25 years.

The equipment's sale resulted in:

- a.** a gain of \$1.4 million
- b. a loss of \$1.7 million
- c. a gain of \$3.8 million
- d. a gain of \$6.2 million

$$\begin{aligned}\text{Gain} &= \text{Cash Received} - \text{Book Value} \\ \text{Gain} &= \$11,000,000 - \$9,600,000 = \\ &\$1,400,000\end{aligned}$$

Cash received		\$11.0
Original cost of cargo plane	\$ 30.0	
Less: Accumulated depreciation (\$1.2 million × 17 years)	<u>(20.4)</u>	
Less: Net book value at date of sale		(9.6)
Gain on sale of cargo plane		<u>\$ 1.4</u>

Disposal of Property, Plant, and Equipment (5 of 5)

Prepare the journal entry to update depreciation expense for Year 17:

	Debit	Credit
Depreciation expense (+E, −SE)	1.2	
Accumulated depreciation (+XA, −A)		1.2

Prepare the journal entry to record the sale:

	Debit	Credit
Cash (+A)	11.0	
Accumulated depreciation (−XA, +A)	20.4	
Aircraft (−A)		30.0
Gain on sale of assets (+R, +SE)		1.4

Assets	=	Liabilities	+	Stockholders' Equity
(1) Accumulated depreciation (+XA) −1.2				Depreciation expense (+E) −1.2
(2) Cash +11.0				Gain on sale of asset (+R) +1.4
Aircraft −30.0				
Accumulated depreciation (−XA) +20.4				

Learning Objective 8-6

8-6 Apply measurement and reporting concepts for intangible assets and natural resources.

Nature of Intangible Assets

- Intangible assets are **increasingly important** resources for organizations.
- An intangible asset has value because of certain **rights** and **privileges** often conferred by law on its owner.
- An intangible asset has **no material or physical substance**. Instead, most intangible assets usually are evidenced by a legal document.

Most common types of intangible assets:

- Goodwill (recognized in a business merger or acquisition)
- Trademarks
- Copyrights
- Technology (computer software and website)
- Patents
- Franchises
- Licenses and operating rights
- Other (customer lists/relationships, noncompete covenants, contracts and agreements)

Intangible assets are recorded **at historical cost only if they have been purchased**. If these assets are developed internally by the company, they are expensed when incurred.

Amortization of Intangible Assets

Upon acquisition of intangible assets, managers determine whether they have definite or indefinite lives:

Definite Life

- Cost is allocated (amortized) using the straight-line method over its useful life. Amortization is a cost allocation process similar to depreciation and depletion.
- Most companies do not estimate a residual value.
- The adjusting entry to record amortization of a patent with an \$800,000 purchase price and legal life of 20 years is as follows:

	Debit	Credit
Amortization expense (+E, -SE)	40,000	
Patents (-A) ¹		40,000

Assets	=	Liabilities	+	Stockholders' Equity
Patents -40,000				Amortization expense (+E) -40,000

Indefinite Life

- Not amortized.
- Asset must be reviewed at least annually for possible impairment of value (first using qualitative factors).
- If impaired, the carrying value is reduced to fair market value.

Common Intangibles (1 of 4)

Goodwill:

- Goodwill is only recorded as an asset when one company buys another business. *Internally developed goodwill (such as developing a favorable reputation) is not reported as an asset.*
- The amount of goodwill equals the purchase price of the business less the fair market value of its net assets (assets minus liabilities).
- Goodwill is not amortized but is reviewed annually for impairment.



Tip

Remember –
Goodwill is only
recorded as an asset
when it is purchased.

Salesforce.com's Purchase of Tableau Software	In Millions
Purchase price	\$ 14,845
Less: Fair value of purchased assets and assumed liabilities:	
Current assets	1,274
Operating lease right-of-use assets and other assets	664
Intangible assets (identifiable technology, customer relationships, and other)	3,252
Liabilities	(1,151)
Net assets, at fair value	<u>4,039</u>
Goodwill	<u><u>\$ 10,806</u></u>

Common Intangibles (2 of 4)

Trademarks :

- A special name, image or slogan identified with a product or company.
- Rarely seen on balance sheets because they are only recorded if purchased.
- Companies often spend millions developing trademarks, but most of those expenditures are recorded as expenses rather than being capitalized as an intangible asset.

Copyrights:

- The exclusive right to publish, use, and sell a literary, musical, or artistic work.
- Legal life is the life of the creator plus 70 years.

Technology:

- Website development: Capitalize the costs of acquiring a domain name and developing graphics.
- Software: Capitalize the direct costs of developing software (coding and testing) after the software is technologically feasible. Amortize any costs related to software that is to be sold or leased to customers. Test the software annually for impairment. *Costs incurred during the preliminary concept phase should be expensed.*

Common Intangibles (3 of 4)

Research & Development Costs:

- Research and development costs **are not capitalized** as an intangible asset due to uncertainty as to future benefits.
- These costs should be expensed and recorded on the income statement.



Tip Do not capitalize research and development expenses—R & D is **NOT** an intangible asset.

Patents:

- A patent is an **exclusive right** granted by the federal government for 20 years for inventions and new processes.
- Owner can use, manufacture, and sell both the subject of the patent and the patent itself.
- If internally developed, only the registration fees and legal costs are capitalized (recorded as an asset).

Common Intangibles (4 of 4)

Franchises:

- Franchises may be granted by the government or a company for a specific period and purpose.
- Franchise agreements are contracts.
 - They usually require an investment by the franchisee; therefore, they should be accounted for as intangible assets.
- The life of the franchise agreement depends on the contract and can be for a single year or indefinite period.

Licenses and operating rights:

- These intangible assets are the permissions to use a product or service according to specific terms and conditions.

Accounting for Software-as-a-Service (SaaS) as a Customer



FINANCIAL ANALYSIS

- The demand for companies for public cloud software applications (SaaS software solutions and cloud data analysis) has grown tremendously in recent years.
- The SaaS market is expected to reach over \$800 billion by 2030.
- SaaS customers usually pay a fixed amount monthly or annually for a subscription-based service.

Do the fees paid result in an intangible asset for the customer?

- Accounting for software-as-a-service arrangements is complex.
- If the **customer has the right to take possession** of the software (a software license), costs incurred by customers are to be **capitalized** and **amortized** over the term of the cloud computing arrangement.
- If the **customer does not take possession** of the software, the arrangement is a service contract.
 - Acquisition and implementation costs are **capitalized** and **amortized**.
 - Subscription fees paid to the SaaS provider are **expensed**.

Acquisition and Depletion of Natural Resources

Natural Resources

- Companies develop raw material and products from natural resources, including mineral deposits such as lithium, gold, iron ore, oil wells, and timber tracts.
- These resources are called “wasting assets” because they are depleted (physically used up).
- When natural resources are acquired or developed, they are recorded at cost. The cost is allocated when the resource is used up. This allocation is called **depletion**.
 - Depletion is the process of allocating a natural resource’s cost over the period of its exploitation.
 - The units-of-production method is often used to compute depletion.
 - The amount of the natural resource depletion is **capitalized** (recorded as inventory). The entry to record depletion is:

Dr. Inventory (+A)	x
Cr. Natural Resource (–A)	x
- Later, when the asset is sold, Cost of Goods Sold (an expense) is recorded.

Learning Objective 8-7

8-7 Explain how the acquisition, use, and disposal of long-lived assets impact cash flows.

Product Long-lived Assets, Depreciation, and Amortization



FOCUS ON CASH FLOWS

The acquisition, sale, depreciation, and amortization of long-lived assets are reflected on a company's cash flow statement. When the indirect method is used, adjustments to net income are made for depreciation expense and gains and losses that relate to sales of long-term assets.

	Effect on Cash Flows
Operating activities (indirect method)	
Net income	\$xxx
Adjusted for:	
Depreciation and amortization expense	+
Gains on sale of long-lived assets	-
Losses on sale of long-lived assets	+
Losses on asset impairment write-downs	+
Investing activities	
Purchase of long-lived assets	-
Sale of long-lived assets	

Production Assets and Depreciation



FOCUS ON CASH FLOWS

Here is a partial statement of cash flows for FedEx Corporation

FEDEx CORPORATION	
Consolidated Statement of Cash Flows (in millions of dollars)	
For the Year Ended May 31, 2013	
Cash Flows from Operating Activities:	
Net income	\$3,972
Adjustments to reconcile net income to cash provided by operating activities:	
Depreciation and amortization	4,176
Other (summarized)	700
Net cash provided by (used in) operating activities	8,848
Cash Flows from Investing Activities:	
Capital expenditures (purchases of property and equipment)	(6,174)
Proceeds from asset dispositions and other	0
Net cash used in investing activities	(6,174)

A Misinterpretation

FINANCIAL ANALYSIS

Some analysts misinterpret the meaning of a noncash expense by saying that “cash is provided by depreciation.”

- Although depreciation is added in the operating section of the statement of cash flows, when using the direct method, **depreciation is not a source of cash.**
- Depreciation expense reduces the reported net income for a company, but it does not reduce the amount of cash generated by the company because it is a **noncash expense.**
- Cash from operations is the cash provided by selling goods and services.
- Although depreciation is a noncash expense, the **depreciation for tax purposes can increase a company's cash flows.** The noncash tax depreciation reduces the taxable income and taxes a company must pay.

Chapter Supplement:

Changes in Depreciation Estimates (1 of 3)

Chapter
Supplement

Depreciation is based on two estimates:

ESTIMATED
useful life

ESTIMATED
residual value

If the estimates change, then the undepreciated asset balance (less any residual value at the date of the change) should be depreciated **over the remaining useful life.**

If improvements are made that extend the assets useful life, the depreciation must also be recalculated.

Chapter Supplement: Changes in Depreciation Estimates (2 of 3)

Chapter Supplement

To compute the new depreciation expense due to a change in estimate, substitute:

- the net book value for the original acquisition cost,
- the new residual value for the original amount, and
- the estimated remaining life in place of the original estimated life.

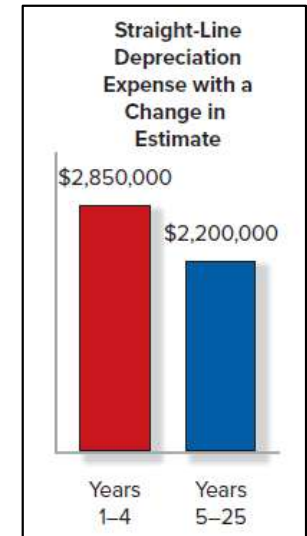
Original Straight-Line Formula Modified for a Change in Estimate:

$(\text{Cost} - \text{Residual Value}) \times 1/\text{Useful Life} = \text{Original Depreciation Expense}$

$(\text{Net Book Value} - \text{New Residual Value}) \times 1/\text{Remaining Life} = \text{Revised Depreciation Expense}$

Chapter Supplement: Changes in Depreciation Estimates (3 of 3)

- FedEx purchased an aircraft for \$60 million with an estimated useful life of 20 years and estimated residual value of \$3 million.
- At the end of year 5, Southwest changed the estimated useful life to 25 years and lowered the residual value to \$2.4 million.
- Calculate depreciation expense for the fifth year using the straight-line method.



Original depreciation expense

$$(\$60,000,000 - \$3,000,000) \times 1/20 = \$ 2,850,000 \text{ per year}$$

$$\times 4 \text{ years}$$

Accumulated depreciation at the end of year 4: \$11,400,000

Net book value at the end of year 4

Acquisition cost	\$60,000,000
Less: Accumulated depreciation	<u>11,400,000</u>
Net book value	<u><u>\$48,600,000</u></u>

Remaining Life

New estimated life	25 years
In the past	<u>4 years</u>
Remaining	<u><u>21 years</u></u>

Depreciation in years 5 through 25 based on changes in estimates

$$(\text{Net Book Value} - \text{New Residual Value}) \times 1/\text{Remaining Years} = \text{New Depreciation Expense}$$

$$(\$48,600,000 - \$2,400,000) \times 1/21 \text{ years} = \underline{\underline{\$2,200,000 \text{ per year}}}$$

Chapter
Supplement